GATE ECE Learning & Assessment Platform

Official Installation, Cloning & Full-Stack Deployment Guide

Target Repository: https://github.com/Rashmitha132/GATE_ECE.git
Technology Stack: FastAPI (Python) • SQLite/PostgreSQL • React 18 • TypeScript • Vite • Tailwind CSS

1. System Prerequisites

Before starting, ensure your operating system has the following core tools installed:

Component	Minimum Version	Verification Command
Git CLI	2.30+ (Git for Windows)	`git --version`
Python Runtime	Python 3.10 - 3.14	`python --version`
Node.js & npm	Node v18.0+ / npm 9.0+	`node -v && npm -v`
Web Browser	Google Chrome / Edge / Firefox	Modern evergreen browser

2. Step 1: Clone the Repository

Open PowerShell, Command Prompt, or your Terminal, navigate to the folder where you want to keep the project, and run:

```
git clone https://github.com/Rashmitha132/GATE_ECE.git
cd GATE_ECE
```

Your project directory structure will look like this:

```
GATE_ECE/
■■■ backend/             # FastAPI REST API, database models, questions bank
■   ■■■ app/             # Application routers, schemas, core security & DB
■   ■■■ requirements.txt # Python dependencies
■   ■■■ .env.example     # Backend environment configuration template
■■■ frontend/            # React + TypeScript + Vite + Tailwind CSS
■   ■■■ src/             # Pages, components, contexts, state
■   ■■■ package.json     # Node.js dependencies
■   ■■■ .env.example     # Frontend API connection template
■■■ README.md            # Project overview documentation
```

3. Step 2: Backend Setup & Execution (FastAPI)

The backend powers the authentication, question bank, test submission evaluation, and analytics.

3.1 Navigate to backend directory:

```
cd backend
```

3.2 Create and activate an isolated Python Virtual Environment:

On **Windows (PowerShell)**:

```
python -m venv venv
.\venv\Scripts\activate
```

Note: If PowerShell shows a script execution error, run: `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned` *once.*

On **macOS / Linux**:

```
python3 -m venv venv
source venv/bin/activate
```

3.3 Install Python packages:

```
pip install -r requirements.txt
```

3.4 Configure Environment Variables (.env):

Create a local `.env` file inside the `backend/` folder by copying the provided template:

On Windows:

```
copy .env.example .env
```

On macOS / Linux:

```
cp .env.example .env
```

Inside `backend/.env`, verify the following default values (pre-configured for local testing):

```
SECRET_KEY=gsss_gate_ece_secure_secret_key_production_2026
SECURITY_SALT=gsss_gate_ece_salt_
ALLOWED_ORIGINS=http://localhost:5173,http://127.0.0.1:5173
DATABASE_URL=sqlite:///./gate_ece.db
```

3.5 Initialize and Seed the Database:

Populate the database with the official syllabus, 1,040 authentic GATE questions, formula sheets, and default credentials:

```
python app/data/seed.py
```

You should see: **`Database seeded successfully with all GATE ECE questions & users!`**

3.6 Start the FastAPI Server:

```
python -m uvicorn app.main:app --reload --host 127.0.0.1 --port 8000
```

Your backend is now live! Verify by opening your browser at:

- **Swagger Interactive API Docs:** http://127.0.0.1:8000/docs
- **Alternative ReDoc Docs:** http://127.0.0.1:8000/redoc

4. Step 3: Frontend Setup & Execution (React + Vite)

Keep your backend terminal window running. Open a **second terminal window** for the frontend.

4.1 Navigate to frontend directory:

```
cd GATE_ECE/frontend
```

4.2 Install npm packages:

```
npm install
```

4.3 Configure Frontend Environment (.env):

Create a `.env` file inside the `frontend/` folder:

On Windows:

```
copy .env.example .env
```

On macOS / Linux:

```
cp .env.example .env
```

Ensure the file contains the local backend URL:

```
VITE_API_URL=http://localhost:8000/api
```

4.4 Start the Vite Development Server:

```
npm run dev
```

Open your web browser and go to:

Web Application URL: http://localhost:5173
The official GSSSIETW Department of ECE Portal landing page will load with the department banner, curriculum options, and portal sign-in.

5. Default Credentials & Portal Roles

The platform features 3 completely isolated portals. Use the seeded credentials below for evaluation:

Portal / Role	Email Address	Password	Authorized Features
Student Portal	student@gsss.edu.in	`password123`	Practice 1,040 questions, Full Mock Exams, Concept Notes, Performance Graph, Formula Sheets
Faculty Portal	faculty@gsss.edu.in	`password123`	Add/Edit Question Bank, Upload study notes (PDFs), Subject coverage monitoring, Student analytics
Administrator	admin@gsssietw.ac.in	`admin123`	Total user management, Department activity metrics, Security audit logs, System settings

6. Key Functional Capabilities

• **1,040 Authentic GATE Questions:** Curated across all 10 GATE ECE subjects (Engineering Mathematics, Networks, Electronic Devices, Analog Circuits, Digital Circuits, Signals & Systems, Control Systems, Communications, Electromagnetics, General Aptitude).

• **GATE Standard Question Types:** Fully supports Multiple Choice Questions (MCQ with negative marking), Multiple Select Questions (MSQ), and Numerical Answer Type (NAT with virtual keypad).

• **GSSSIETW Institutional Branding:** Full college banner with ISO/NAAC/NBA accreditation badges, departmental color accents, and responsive layout.

• **Curriculum & Syllabus Viewer:** Interactive subject breakdown with downloadable official GATE syllabus PDFs.

• **Study Resources Repository:** Built-in formula sheets, Sedra & Smith analog digests, NPTEL syllabus roadmaps, and solved previous year papers (2023 & 2024).

• **Department Notifications:** Interactive bell modal with unread indicators and bulk 'Mark all read' action.

7. Troubleshooting & FAQ

Q: PowerShell 'execution of scripts is disabled'

A: Run `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned` in PowerShell as Administrator, then reactivate the venv.

Q: Port 8000 or Port 5173 already in use

A: Another instance is running. Stop previous terminal windows with `Ctrl + C` or kill Python/Node via Task Manager.

Q: Database locked error in SQLite

A: Only one process should write to SQLite at a time. Ensure only one FastAPI terminal instance is running.

Q: Need to reset passwords or questions

A: Simply re-run `python app/data/seed.py` from inside the `backend/` folder to reset the database to a clean baseline.

Need Support? Department of Electronics & Communication Engineering
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Official Website: https://www.gsssietw.ac.in | GitHub: https://github.com/Rashmitha132/GATE_ECE